Le Nguyen Cong Danh | Hardware Engineer

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Summary

Dedicated and detail-oriented hardware engineer specializing in IC design and IoT systems. Strong foundation in microelectronics with hands-on experience in circuit design, optimization, and embedded system integration. Committed to developing innovative and reliable hardware solutions for real-world applications.

Education

Ho Chi Minh City University of Technology and Education
Major in Microelectronics Engineering — *2021 - Present*

Work Experience

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Currently seeking industry experience to further develop practical skills in microelectronics and embedded systems.

Personal Projects

Smart Home Fire Warning IoT System:
- Designed and implemented a real-time IoT system for fire detection and warning in residential environments.
- Utilized sensors, microcontrollers, and wireless modules for efficient and low-power operation.
- Integrated mobile notification feature to alert homeowners instantly.

PLL Optimization in IC Design:
- Researched and optimized Phase-Locked Loop (PLL) circuits for better stability and low jitter.
- Conducted simulations and layout using EDA tools for silicon-proven performance.
- Enhanced power efficiency and minimized phase noise for high-frequency applications.

Body Temperature Screening System (PIC 16F887):
- Designed a body temperature screening system using PIC 16F887 microcontroller.
- Developed firmware to read temperature data from sensors and trigger alerts when abnormal values are detected.
- Deployed system with LCD display and buzzer notification for real-time monitoring in public areas.

Distributed Antenna System (DAS) for Buildings:
- Developed a model for indoor DAS to improve wireless coverage within multi-story buildings.
- Simulated RF signal distribution and antenna placement using specialized tools.
- Proposed layout and cable routing plan for optimal signal strength in large-area coverage.

FM Modulator and Demodulator Design:
- Designed and implemented FM modulation and demodulation circuits using analog components.
- Analyzed frequency deviation, bandwidth, and noise immunity through practical testing.
- Applied the system in basic audio transmission scenarios for academic demonstration.

Semi-Active Radar System Design:
- Developed a conceptual radar system using semi-active principles for target detection.
- Designed signal processing blocks for RF signal transmission and echo analysis.
- Simulated radar parameters and detection accuracy using MATLAB.

Drone Research and Development:
- Built and tested custom drone models focusing on flight stability and payload capacity.
- Integrated sensors (IMU, GPS) and microcontrollers for autonomous flight control.
- Explored control algorithms and wireless telemetry systems for long-range operation.

Skills

Hardware Design: Analog & Digital Circuits, PCB Design, Microcontroller Programming — *Proficient*
Tools: Altium Designer, Proteus, Cadence Virtuoso, Verilog, MATLAB, Multisim — *Proficient*
Embedded Systems: Arduino, ESP32, STM32, IoT Protocols (MQTT, HTTP) — *Experienced*
Soft Skills: Problem-Solving, Team Collaboration, Communication — *Strong*

Achievements

Expert Challenge Season 4: First Prize Winner

University Research Award: First Prize - Scientific Research Competition at University Level

Robocon University Level: Participated in internal Robocon competition, gaining practical experience in embedded robotics